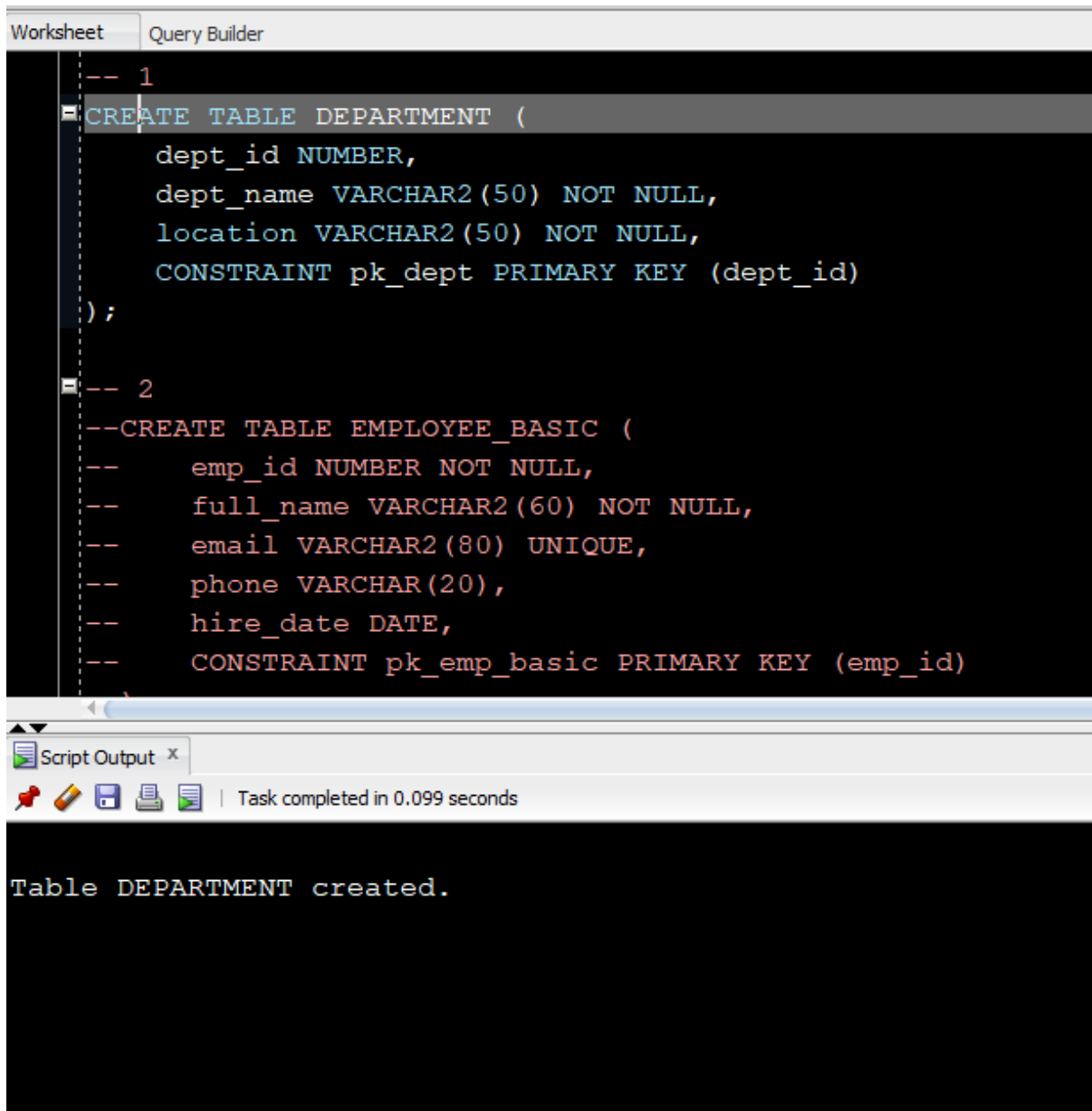


DBS Lab Task 3

1)



```
-- 1
CREATE TABLE DEPARTMENT (
    dept_id NUMBER,
    dept_name VARCHAR2(50) NOT NULL,
    location VARCHAR2(50) NOT NULL,
    CONSTRAINT pk_dept PRIMARY KEY (dept_id)
);

-- 2
--CREATE TABLE EMPLOYEE_BASIC (
--    emp_id NUMBER NOT NULL,
--    full_name VARCHAR2(60) NOT NULL,
--    email VARCHAR2(80) UNIQUE,
--    phone VARCHAR(20),
--    hire_date DATE,
--    CONSTRAINT pk_emp_basic PRIMARY KEY (emp_id)
```

Script Output x

Task completed in 0.099 seconds

Table DEPARTMENT created.

2)

```
-- 2
CREATE TABLE EMPLOYEE_BASIC (
    emp_id NUMBER NOT NULL,
    full_name VARCHAR2(60) NOT NULL,
    email VARCHAR2(80) UNIQUE,
    phone VARCHAR(20),
    hire_date DATE,
    CONSTRAINT pk_emp_basic PRIMARY KEY (emp_id)
);

----- 3
--CREATE TABLE JOB_ROLE (
```

Script Output x

Task completed in 0.024 seconds

Table DEPARTMENT created.

Table EMPLOYEE_BASIC created.

3)

```
-- 3
CREATE TABLE JOB_ROLE (
    job_id NUMBER,
    job_title VARCHAR2(40),
    min_salary NUMBER DEFAULT 20000,
    max_salary NUMBER,
    CONSTRAINT pk_job_role PRIMARY KEY (job_id),
    CONSTRAINT chk_job_sal CHECK(max_salary > min_salary)
);

----- 4
```

Script Output x

Task completed in 0.042 seconds

Table DEPARTMENT created.

Table EMPLOYEE_BASIC created.

Table JOB_ROLE created.

4)

```
29 -- 4
30 CREATE TABLE EMPLOYEE_PERSONAL (
31     emp_id NUMBER,
32     gender CHAR(1),
33     date_of_birth DATE,
34     marital_status VARCHAR2(10),
35     CONSTRAINT pk_emp_per PRIMARY KEY (emp_id),
36     CONSTRAINT chk_gender CHECK (gender IN ('M', 'F')),
37     CONSTRAINT chk_marriage_status CHECK(marital_status IN ('SINGLE', 'MARRIED'))
38 );
39
40 ----- 5
41 --CREATE TABLE EMPLOYEE_WORK (
```

Script Output x

Task completed in 0.034 seconds

Table EMPLOYEE_PERSONAL created.

5)

```
39
40 -- 5
41 CREATE TABLE EMPLOYEE_WORK (
42     emp_id NUMBER,
43     dept_id NUMBER,
44     job_id NUMBER,
45     joining_date DATE,
46     CONSTRAINT fk_dept FOREIGN KEY (dept_id) REFERENCES DEPARTMENT(dept_id),
47     CONSTRAINT fk_job FOREIGN KEY (job_id) REFERENCES JOB_ROLE(job_id),
48     CONSTRAINT pk_emp_wrk PRIMARY KEY (emp_id)
49 );
50
51 ----- 6
```

Script Output x

Task completed in 0.036 seconds

Table EMPLOYEE_PERSONAL created.

Table EMPLOYEE_WORK created.

6)

```
50
51 -- 6
52 CREATE TABLE PROJECT (
53     project_id NUMBER,
54     project_name VARCHAR2(60),
55     start_date DATE,
56     end_date DATE,
57     CONSTRAINT pk_proj_id PRIMARY KEY (project_id),
58     CONSTRAINT chk_proj_date CHECK (start_date < end_date)
59 );
60
61 ----- 7
62 --CREATE TABLE PROJECT_ASSIGNMENT (
```

Script Output x

Task completed in 0.035 seconds

Table EMPLOYEE_PERSONAL created.

Table EMPLOYEE_WORK created.

Table PROJECT created.

7)

```

61  -- 7
62  CREATE TABLE PROJECT_ASSIGNMENT (
63      emp_id NUMBER,
64      project_id NUMBER,
65      assigned_date DATE,
66      CONSTRAINT pk_emp_proj_id PRIMARY KEY (emp_id, project_id),
67      CONSTRAINT fk_emp_id FOREIGN KEY (emp_id) REFERENCES EMPLOYEE_WORK(emp_id),
68      CONSTRAINT fk_proj_id FOREIGN KEY (project_id) REFERENCES PROJECT(project_id)
69  );
70
71  ---- 8

```

Script Output x
 Task completed in 0.035 seconds

Table EMPLOYEE_PERSONAL created.

Table EMPLOYEE_WORK created.

Table PROJECT created.

Table PROJECT_ASSIGNMENT created.

8)

```

71  -- 8
72  ALTER TABLE PROJECT_ASSIGNMENT DROP CONSTRAINT fk_emp_id;
73  ALTER TABLE PROJECT_ASSIGNMENT ADD CONSTRAINT fk_emp_id
74      FOREIGN KEY (emp_id) REFERENCES EMPLOYEE_WORK(emp_id)
75      ON DELETE CASCADE;
76
77  ---- 9

```

Script Output x
 Task completed in 0.044 seconds

Table EMPLOYEE_WORK created.

Table PROJECT created.

Table PROJECT_ASSIGNMENT created.

Table PROJECT_ASSIGNMENT altered.

Table PROJECT_ASSIGNMENT altered.

9)

```
77 -- 9
78 ALTER TABLE EMPLOYEE_WORK DROP CONSTRAINT fk_dept;
79 ALTER TABLE EMPLOYEE_WORK ADD CONSTRAINT fk_dept
80     FOREIGN KEY (dept_id) REFERENCES DEPARTMENT(dept_id)
81     ON DELETE SET NULL;
82
83 ----- 10
84 --CREATE TABLE CLIENT (
85 --    client_id NUMBER,
86 --    client_name VARCHAR2(40),
87 --    email VARCHAR2(60),
```

Script Output x

Task completed in 0.037 seconds

Table EMPLOYEE_WORK altered.

Table EMPLOYEE_WORK altered.

10)

```
82
83 -- 10
84 CREATE TABLE CLIENT (
85     client_id NUMBER,
86     client_name VARCHAR2(40),
87     email VARCHAR2(60),
88     client_type VARCHAR2(20),
89     country VARCHAR2(30)
90 );
91
92 ----- 11
```

Script Output x

Task completed in 0.024 seconds

```
Table EMPLOYEE_WORK altered.

Table EMPLOYEE_WORK altered.

Table CLIENT created.
```

11)

```
91
92 -- 11
93 ALTER TABLE CLIENT ADD CONSTRAINT pk_cl_id PRIMARY KEY (client_id);
94 ALTER TABLE CLIENT ADD CONSTRAINT email_uni UNIQUE (email);
95 ALTER TABLE CLIENT ADD CONSTRAINT cli_type
96     CHECK (client_type IN ('LOCAL', 'INTERNATIONAL'));
97
98 ----- 12
```

Script Output x

Task completed in 0.043 seconds

Table EMPLOYEE_WORK altered.

Table CLIENT created.

Table CLIENT altered.

Table CLIENT altered.

Table CLIENT altered.

12)

```
98 -- 12
99 ALTER TABLE CLIENT MODIFY client_name VARCHAR2(80);
100 ALTER TABLE CLIENT MODIFY email VARCHAR2(60) NOT NULL;
101 ALTER TABLE CLIENT MODIFY country VARCHAR2(30) DEFAULT 'PAKISTAN';
102
103 ----- 13
```

Script Output x

Task completed in 0.066 seconds

Table CLIENT altered.

Table CLIENT altered.

Table CLIENT altered.

Table CLIENT altered.

Table CLIENT altered.

13)


```
103 -- 13
104 ALTER TABLE CLIENT DROP CONSTRAINT cli_type;
105 ALTER TABLE CLIENT RENAME CONSTRAINT email_uni TO uniq_email;
106
107 ----- 14
108 --CREATE TABLE EMPLOYEE_BACKUP AS SELECT * FROM EMPLOYEE_WORK;
```

Script Output x

Task completed in 0.04 seconds

Table CLIENT altered.

Table CLIENT altered.

14)

```
106
107 -- 14
108 CREATE TABLE EMPLOYEE_BACKUP AS SELECT * FROM EMPLOYEE_WORK;
109
110 ----- 15
111 --CREATE TABLE PAYROLL (
112 --    emp_id NUMBER,
113 --    salary NUMBER,
114 --    bonus NUMBER DEFAULT 0,
115 --    pay_date DATE,
```

Script Output x

Task completed in 0.023 seconds

Table EMPLOYEE_BACKUP created.

15)

```
109
110 -- 15
111 CREATE TABLE PAYROLL (
112     emp_id NUMBER,
113     salary NUMBER,
114     bonus NUMBER DEFAULT 0,
115     pay_date DATE,
116     CONSTRAINT pk_emp_paydate PRIMARY KEY (emp_id, pay_date),
117     CONSTRAINT chk_pos_nums CHECK (emp_id >= 0 AND salary >= 0 AND bonus >= 0),
118     CONSTRAINT fk_emp_id_payroll FOREIGN KEY (emp_id) REFERENCES EMPLOYEE_WORK(emp_id)
119 );
120
121 ---- 16
```

Script Output x

Task completed in 0.064 seconds

Table PAYROLL created.

16)

```
121 -- 16
122 CREATE TABLE ATTENDANCE (
123     attendance_id NUMBER,
124     emp_id NUMBER,
125     project_id NUMBER,
126     dept_id NUMBER,
127     attendance_date DATE,
128     CONSTRAINT pk_attend_id PRIMARY KEY (attendance_id),
129     CONSTRAINT fk_emp_id_attend FOREIGN KEY (emp_id) REFERENCES EMPLOYEE_WORK(emp_id),
130     CONSTRAINT fk_proj_id_attend FOREIGN KEY (project_id) REFERENCES PROJECT(project_id),
131     CONSTRAINT fk_dept_id_attend FOREIGN KEY (dept_id) REFERENCES DEPARTMENT(dept_id)
132 );
133
134 ---- 17
135 --ALTER TABLE PROJECT_ASSIGNMENT DISABLE CONSTRAINT fk_emp_id;
136 --ALTER TABLE PROJECT_ASSIGNMENT ENABLE CONSTRAINT fk_emp_id;
```

Script Output x

Task completed in 0.056 seconds

Table PAYROLL created.

Table ATTENDANCE created.

17)

```
133 |
134 | -- 17
135 | ALTER TABLE PROJECT_ASSIGNMENT DISABLE CONSTRAINT fk_emp_id;
136 | ALTER TABLE PROJECT_ASSIGNMENT ENABLE CONSTRAINT fk_emp_id;
137 |
138 | ---- 18
139 | --CREATE TABLE LEAVE_APPLICATION (
```

Script Output x

Task completed in 0.06 seconds

Table PAYROLL created.

Table ATTENDANCE created.

Table PROJECT_ASSIGNMENT altered.

Table PROJECT_ASSIGNMENT altered.

18)

```
137
138  -- 18
139 CREATE TABLE LEAVE_APPLICATION (
140     leave_id NUMBER,
141     emp_id NUMBER,
142     start_date DATE,
143     end_date DATE,
144     leave_days NUMBER,
145     CONSTRAINT pk_leave_id PRIMARY KEY (leave_id),
146     CONSTRAINT fk_emp_id_leave FOREIGN KEY (emp_id) REFERENCES EMPLOYEE_WORK(emp_id),
147     CONSTRAINT chk_leave_days CHECK (leave_days <= 30),
148     CONSTRAINT chk_date_logic CHECK (start_date < end_date)
149 );
150
151 ---- 19
```

Script Output x

Task completed in 0.058 seconds

Table PAYROLL created.

Table ATTENDANCE created.

Table PROJECT_ASSIGNMENT altered.

Table PROJECT_ASSIGNMENT altered.

Table LEAVE_APPLICATION created.

19)

```
150
151 -- 19
152 DROP TABLE DEPARTMENT;
153
154 ---- 20
155
```

Script Output x

Task completed in 0.047 seconds

Table LEAVE_APPLICATION created.

Error starting at line : 152 in command -
DROP TABLE DEPARTMENT

Error report -

ORA-02449: unique/primary keys in table referenced by foreign keys
02449. 00000 - "unique/primary keys in table referenced by foreign keys"

*Cause: An attempt was made to drop a table with unique or
primary keys referenced by foreign keys in another table.

*Action: Before performing the above operations the table, drop the
foreign key constraints in other tables. You can see what
constraints are referencing a table by issuing the following
command:
SELECT * FROM USER_CONSTRAINTS WHERE TABLE_NAME = "tabnam";

20)